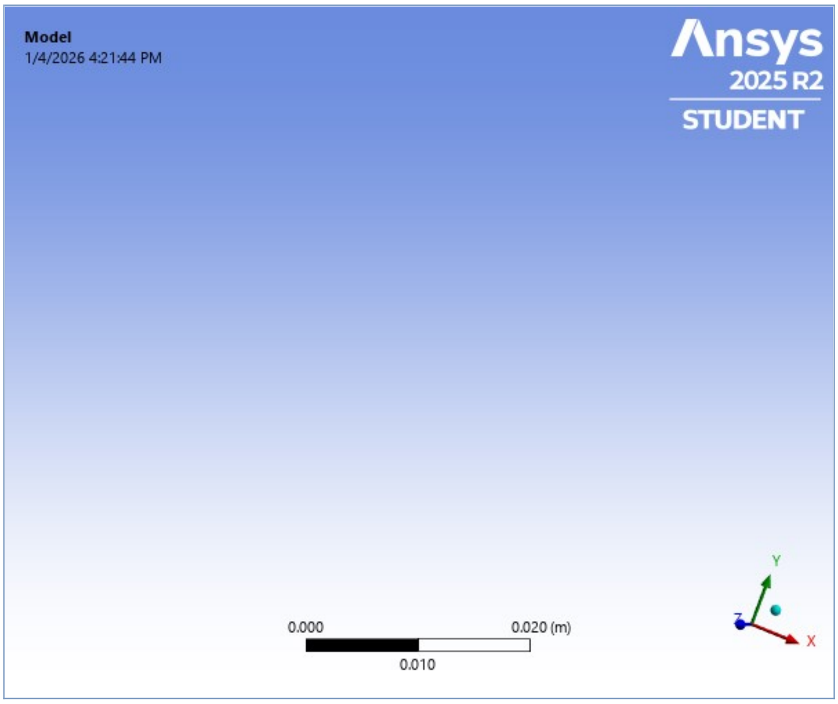
Project*

First Saved	Saturday, March 1, 2025
Last Saved	Saturday, March 1, 2025
Product Version	2025 R1
Save Project Before Solution	No
Save Project After Solution	No



Contents

Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

FIGURE 1
Model (A4) > Figure

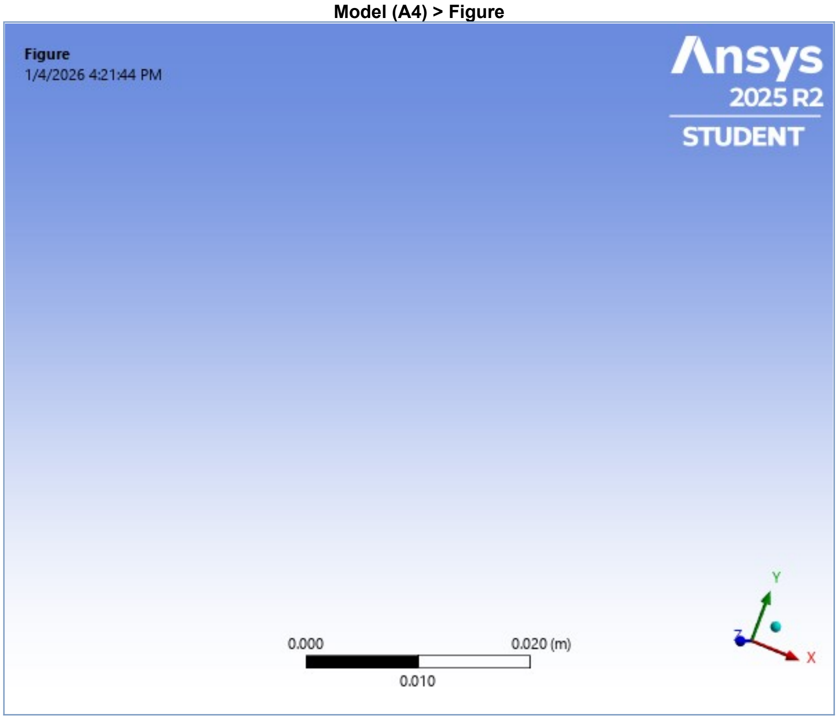


TABLE 2
Model (A4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	*Geometry Import (A3)*
State	Solved
Definition	
Source	E:\from mhmd LAB\Ansys mechanical\1d element lec12\lec12_2\1d elements_files\dp0\SYS\DM\SYS.agdb
Type	DesignModeler
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Advanced Geometry Options	
Compare Parts On Update	No
Analysis Type	3-D

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	E:\from mhmd LAB\Ansys mechanical\1d element lec12\lec12_2\1d elements_files\dp0\SYS\DM\SYS.agdb
Type	DesignModeler
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	6.e-002 m
Length Y	0. m
Length Z	0. m
Properties	
Volume	5.6742e-005 m³
Mass	0.44543 kg
Scale Factor Value	1.
Statistics	
Bodies	3
Active Bodies	3
Nodes	45
Elements	22
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Body Groups

Object Name	*Part*
State	Meshed
Graphics Properties	
Visible	Yes
Definition	
Suppressed	No
Assignment	Structural Steel
Coordinate System	Default Coordinate System
Bounding Box	
Length X	6.e-002 m
Length Y	0. m
Length Z	0. m
Properties	
Volume	5.6742e-005 m³
Mass	0.44543 kg
Statistics	
Nodes	45
Elements	22
Mesh Metric	None
CAD Attributes	
DMBodyGroup	1

TABLE 6
Model (A4) > Geometry > Part > Parts

Object Name	*Line Body*	*Line Body*	*Line Body*
State	Meshed		
Graphics Properties			
Visible	Yes		
Transparency	1		
Definition			
Suppressed	No		
Model Type	Beam		
Stiffness Behavior	Flexible		
Coordinate System	Default Coordinate System		
Reference Temperature	By Environment		
Cross Section	Circular1	Circular2	Circular3
Offset Mode	Refresh on Update		
Offset Type	Centroid		
Treatment	None		
Material			

Assignment	Structural Steel		
Nonlinear Effects	Yes		
Thermal Strain Effects	Yes		
Bounding Box			
Length X	3.e-002 m	2.e-002 m	1.e-002 m
Length Y	0. m		
Length Z	0. m		
Properties			
Volume	3.7697e-005 m³	1.4136e-005 m³	4.9085e-006 m³
Mass	0.29592 kg	0.11097 kg	3.8532e-002 kg
Length	3.e-002 m	2.e-002 m	1.e-002 m
Cross Section Area	1.2566e-003 m²	7.0682e-004 m²	4.9085e-004 m²
Cross Section IYY	1.2565e-007 m²·m²	3.9756e-008 m²·m²	1.9172e-008 m²·m²
Cross Section IZZ	1.2565e-007 m²·m²	3.9756e-008 m²·m²	1.9172e-008 m²·m²
Statistics			
Nodes	23	15	9
Elements	11	7	4
Mesh Metric	None		

FIGURE 2
Model (A4) > Geometry > Figure

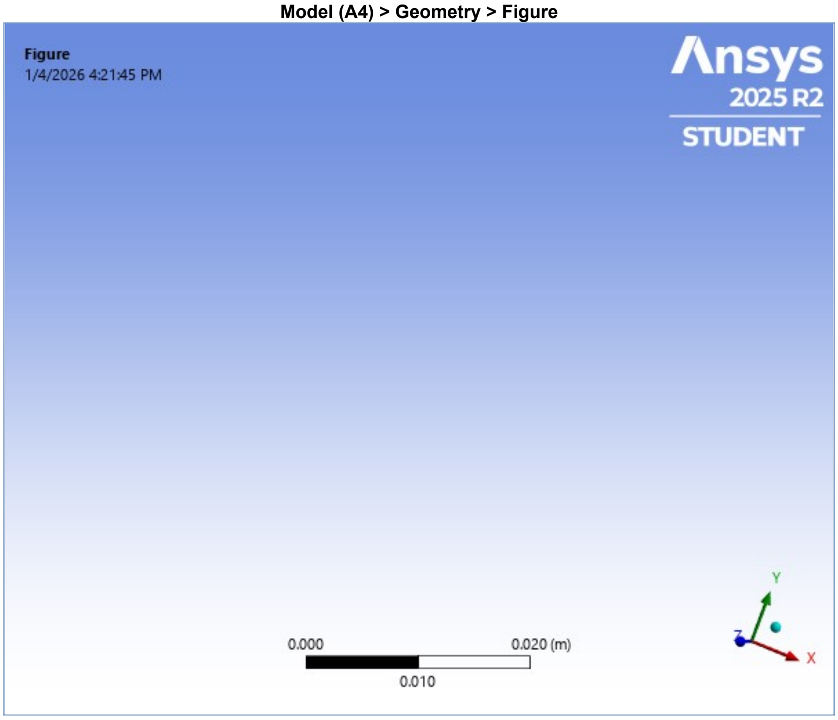


TABLE 7
Model (A4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	1
Material Assignments	0

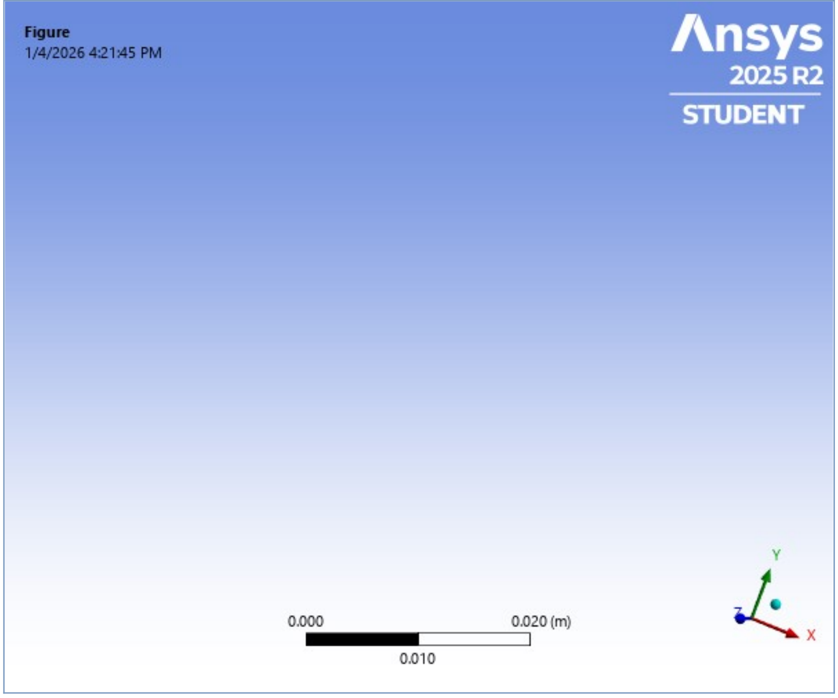
TABLE 8
Model (A4) > Cross Sections

Object Name	*Cross Sections*
State	Fully Defined
Statistics	
Cross Sections	3

TABLE 9
Model (A4) > Cross Sections > Circular1

Object Name	*Circular1*	*Circular2*	*Circular3*
State	Fully Defined		
Definition			
Type	CSOLID		
Import Type	Imported		
Dimensions			
R	2.e-002 m	1.5e-002 m	1.25e-002 m
Physical Properties			
Beam Section	Circular1	Circular2	Circular3
A	1.2566e-003 m²	7.0682e-004 m²	4.9085e-004 m²
Iyy	1.2565e-007 m²·m²	3.9756e-008 m²·m²	1.9172e-008 m²·m²
Izz	1.2565e-007 m²·m²	3.9756e-008 m²·m²	1.9172e-008 m²·m²

FIGURE 3
Model (A4) > Cross Sections > Figure



Coordinate Systems

TABLE 10
Model (A4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. m
Origin Y	0. m
Origin Z	0. m
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Connections

TABLE 11
Model (A4) > Connections

Object Name	*Connections*
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	0
Active Contacts	0
Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

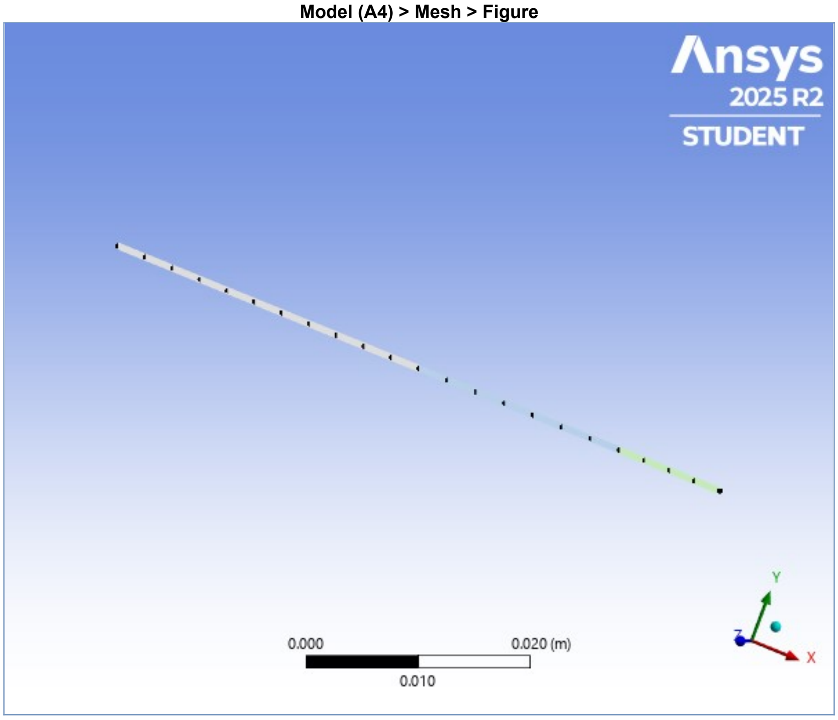
Mesh

TABLE 12
Model (A4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast

Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	6.e-002 m
Average Surface Area	0.0 m²
Minimum Edge Length	1.e-002 m
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	45
Elements	22
Show Detailed Statistics	No

FIGURE 4
Model (A4) > Mesh > Figure



Static Structural (A5)

TABLE 13
Model (A4) > Analysis

Object Name	*Static Structural (A5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 14
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	

Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	E:\from mhmd LAB\Ansys mechanical\1d element lec12\lec12_2\1d elements_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	mks

TABLE 15
Model (A4) > Static Structural (A5) > Loads

Object Name	*Fixed Support*	*Force*
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	1 Vertex	
Definition		
Type	Fixed Support	Force
Suppressed	No	
Define By		Components
Coordinate System		Global Coordinate System
X Component		1.e+005 N (ramped)
Y Component		0. N (ramped)
Z Component		0. N (ramped)

FIGURE 5
Model (A4) > Static Structural (A5) > Fixed Support > Figure

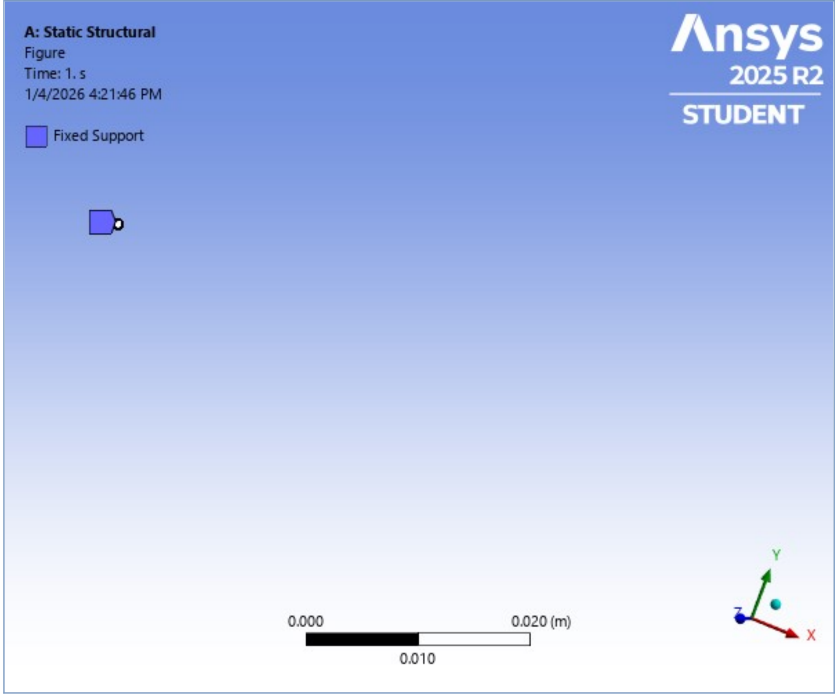


FIGURE 6
Model (A4) > Static Structural (A5) > Force

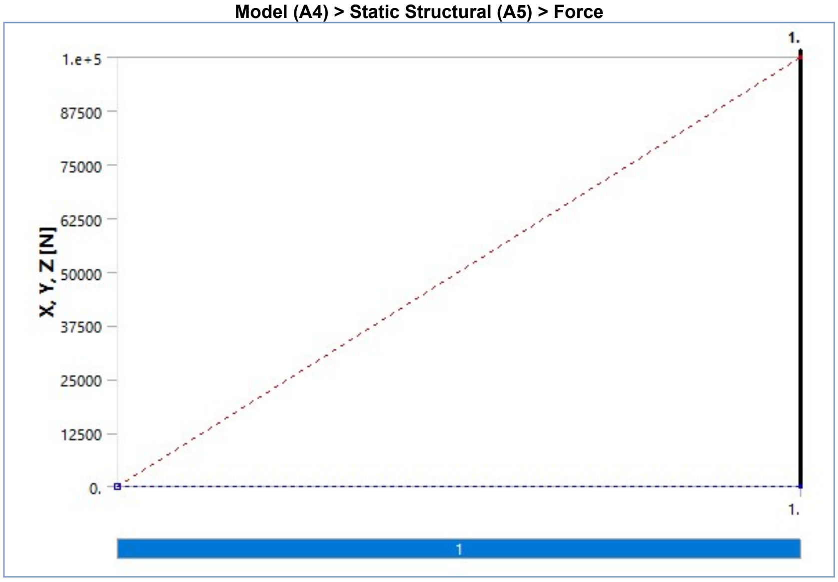
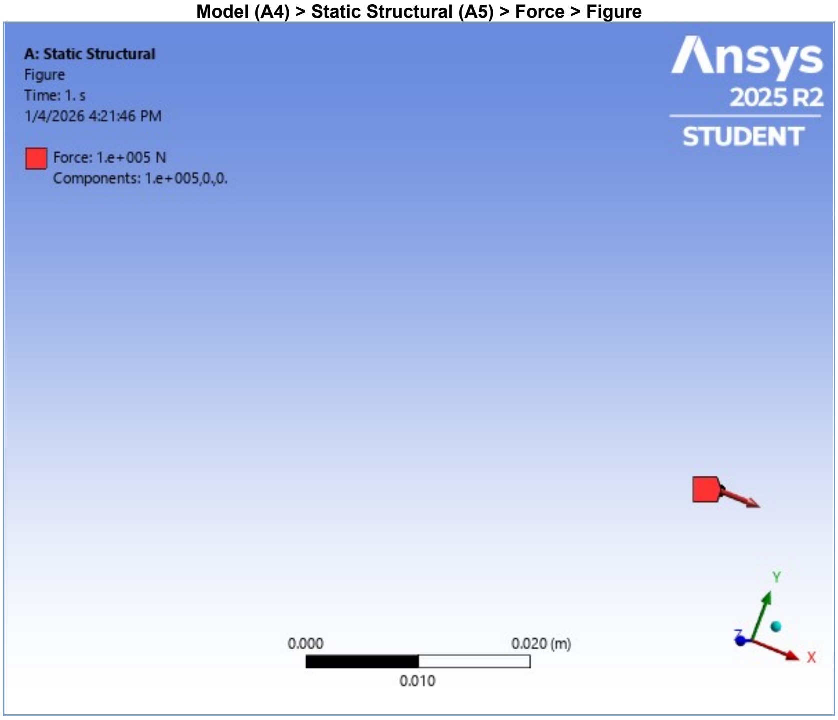


FIGURE 7
Model (A4) > Static Structural (A5) > Force > Figure



Solution (A6)

TABLE 16
Model (A4) > Static Structural (A5) > Solution

Object Name	*Solution (A6)*

State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	2. s
MAPDL Memory Used	188. MB
MAPDL Result File Size	384. KB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 17
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 18
Model (A4) > Static Structural (A5) > Solution (A6) > Results

Object Name	*Total Deformation*
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Total Deformation
By	Time
Display Time	Last
Separate Data by Entity	No
Calculate Time History	Yes
Identifier	
Suppressed	No
Results	
Minimum	0. m
Maximum	3.6298e-005 m
Average	1.4862e-005 m
Minimum Occurs On	Line Body
Maximum Occurs On	Line Body
Information	
Time	1. s
Load Step	1
Substep	1
Iteration Number	1

FIGURE 8
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

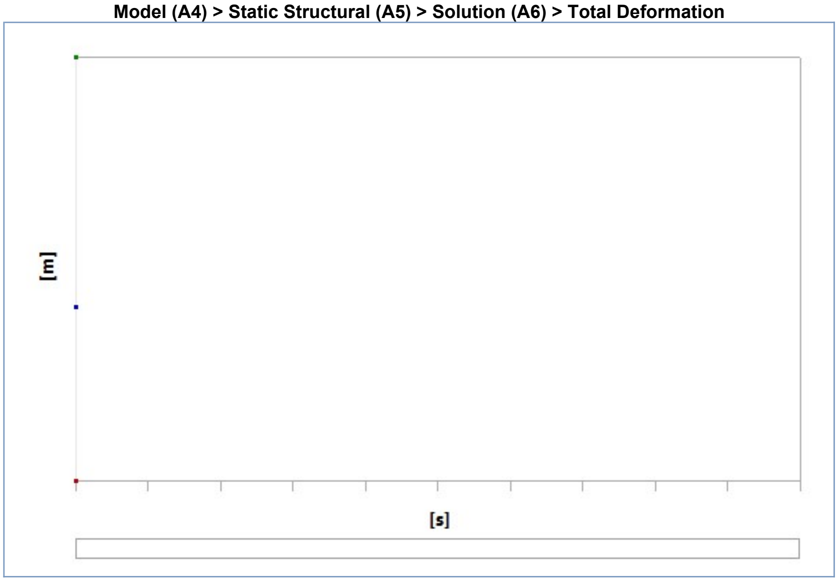


TABLE 19
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [m]	Maximum [m]	Average [m]
1.	0.	3.6298e-005	1.4862e-005

FIGURE 9
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation > Image

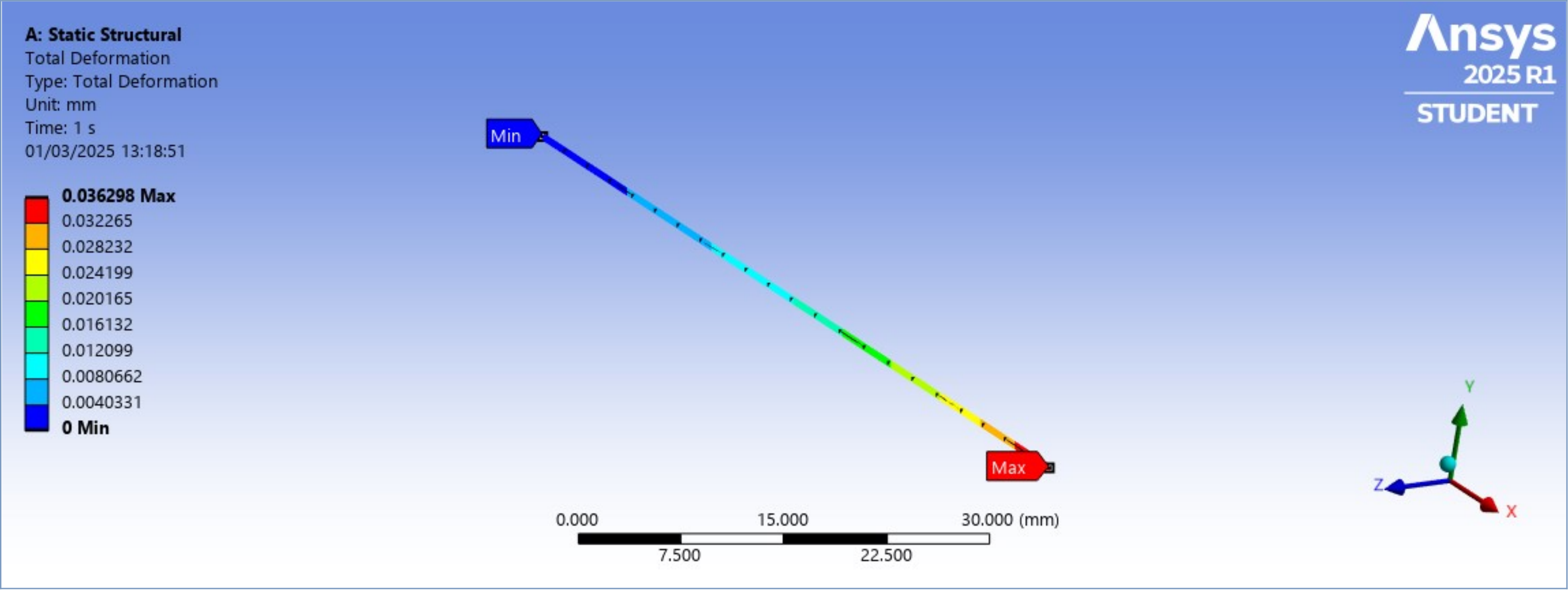


FIGURE 10
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation > Figure

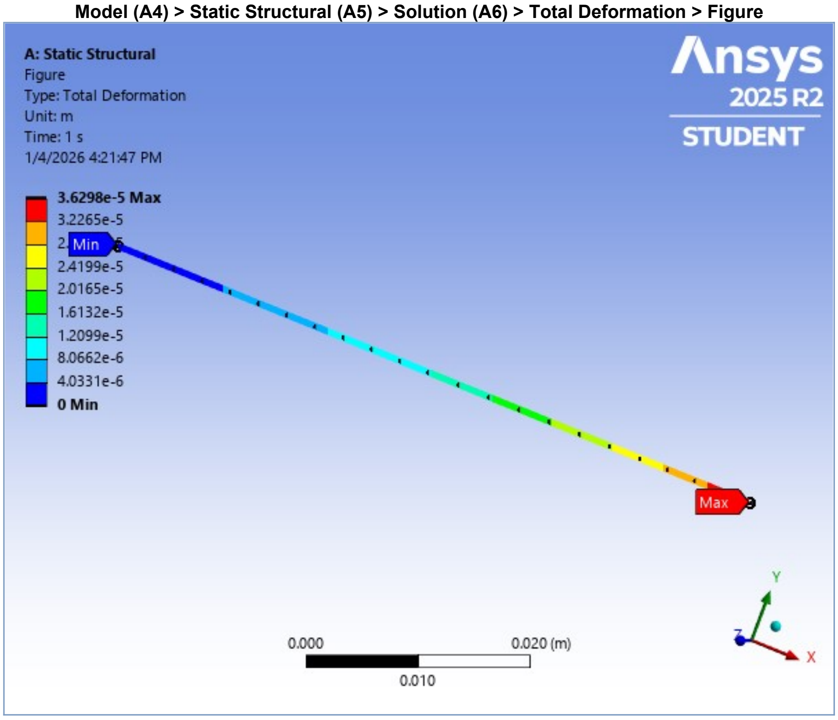


TABLE 20
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool

Object Name	*Beam Tool*	*Beam Tool 2*	*Beam Tool 3*
State	Solved		
Scope			
Geometry	1 Edge		

TABLE 21
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool > Results

Object Name	*Direct Stress*	*Minimum Combined Stress*	*Maximum Combined Stress*
State	Solved		
Definition			
Type	Direct Stress	Minimum Combined Stress	Maximum Combined Stress
By	Time		
Display Time	Last		
Separate Data by Entity	No		
Calculate Time History	Yes		
Identifier			
Suppressed	No		
Integration Point Results			
Display Option	Averaged		
Results			
Minimum	7.9639e+007 Pa		
Maximum	7.9639e+007 Pa		
Average	7.9639e+007 Pa		
Minimum Occurs On	Line Body		
Maximum Occurs On	Line Body		
Information			
Time	1. s		
Load Step	1		
Substep	1		
Iteration Number	1		

FIGURE 11
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool > Direct Stress

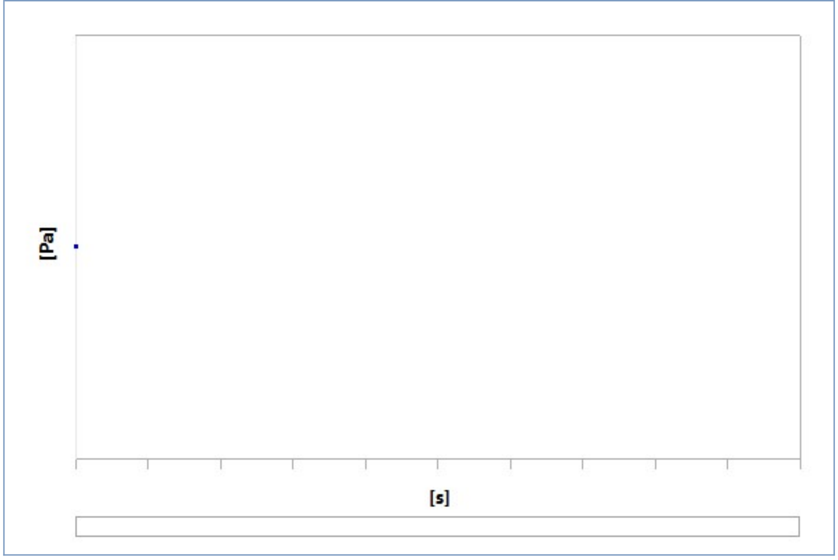


TABLE 22
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool > Direct Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	7.9639e+007	7.9639e+007	7.9639e+007

FIGURE 12
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool > Minimum Combined Stress

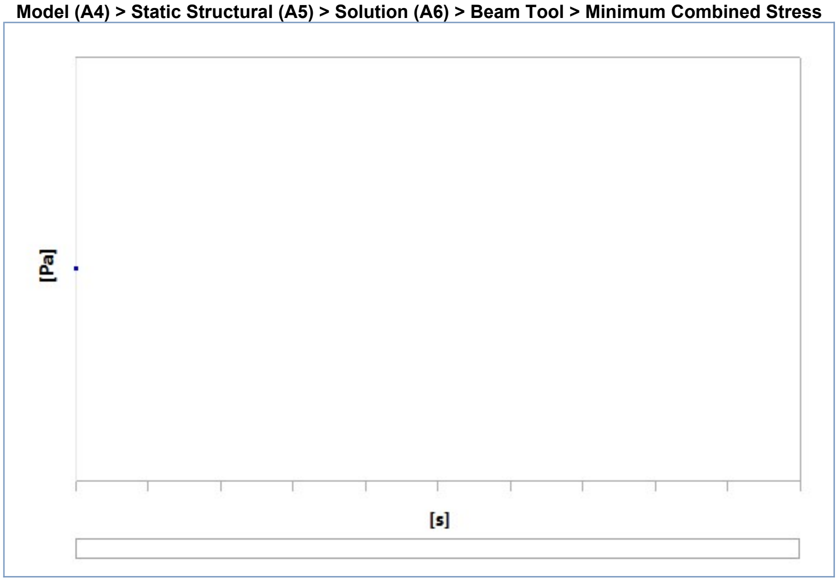


TABLE 23
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool > Minimum Combined Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	7.9639e+007	7.9639e+007	7.9639e+007

FIGURE 13
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool > Maximum Combined Stress

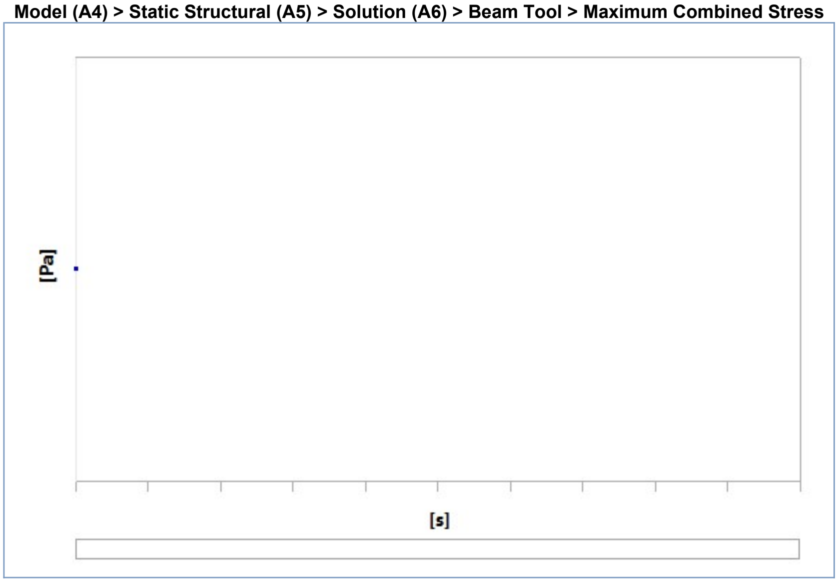


TABLE 24
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool > Maximum Combined Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	7.9639e+007	7.9639e+007	7.9639e+007

TABLE 25
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 2 > Results

Object Name	*Direct Stress*	*Minimum Combined Stress*	*Maximum Combined Stress*

State	Solved		
Definition			
Type	Direct Stress	Minimum Combined Stress	Maximum Combined Stress
By	Time		
Display Time	Last		
Separate Data by Entity	No		
Calculate Time History	Yes		
Identifier			
Suppressed	No		
Integration Point Results			
Display Option	Averaged		
Results			
Minimum	1.4158e+008 Pa		
Maximum	1.4158e+008 Pa		
Average	1.4158e+008 Pa		
Minimum Occurs On	Line Body		
Maximum Occurs On	Line Body		
Information			
Time	1. s		
Load Step	1		
Substep	1		
Iteration Number	1		

FIGURE 14
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 2 > Direct Stress

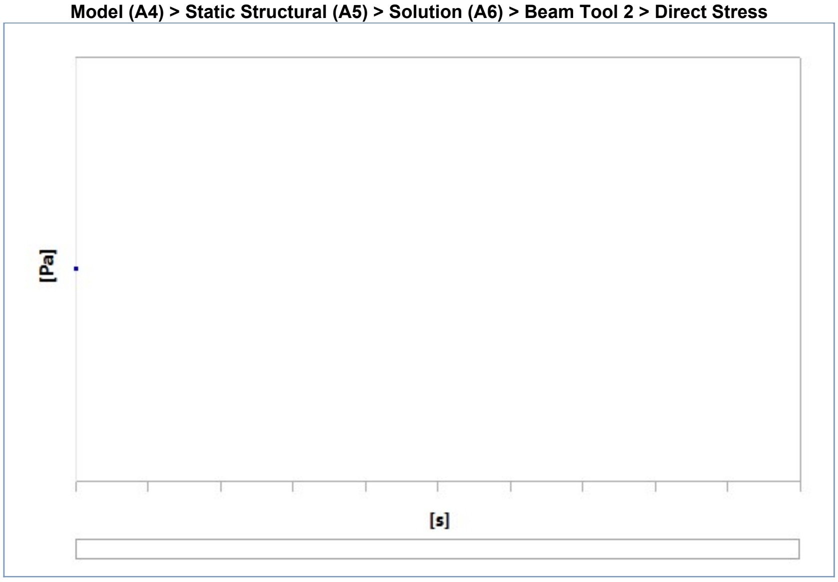


TABLE 26
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 2 > Direct Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	1.4158e+008	1.4158e+008	1.4158e+008

FIGURE 15
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 2 > Minimum Combined Stress

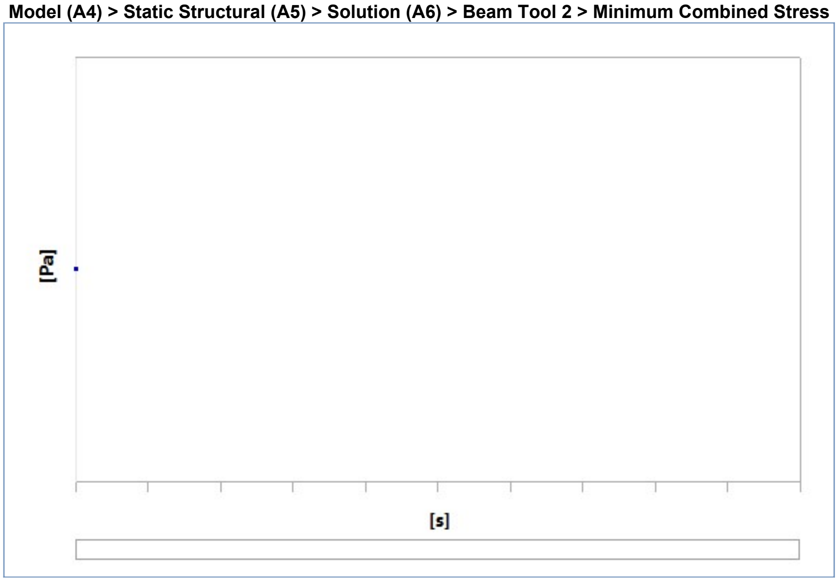


TABLE 27
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 2 > Minimum Combined Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	1.4158e+008	1.4158e+008	1.4158e+008

FIGURE 16
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 2 > Maximum Combined Stress

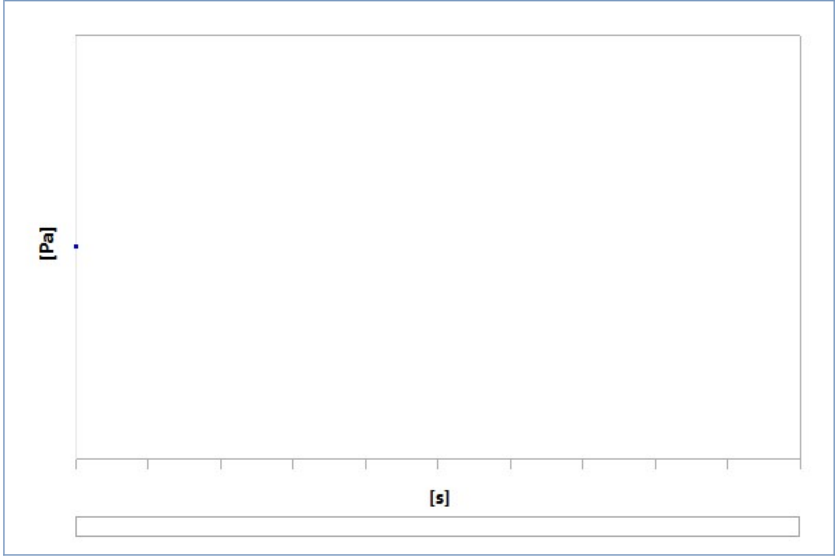


TABLE 28
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 2 > Maximum Combined Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	1.4158e+008	1.4158e+008	1.4158e+008

TABLE 29
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 3 > Results

Object Name	*Direct Stress*	*Minimum Combined Stress*	*Maximum Combined Stress*
State	Solved		
Definition			
Type	Direct Stress	Minimum Combined Stress	Maximum Combined Stress
By	Time		
Display Time	Last		
Separate Data by Entity	No		
Calculate Time History	Yes		
Identifier			
Suppressed	No		
Integration Point Results			
Display Option	Averaged		
Results			
Minimum	2.0388e+008 Pa		
Maximum	2.0388e+008 Pa		
Average	2.0388e+008 Pa		
Minimum Occurs On	Line Body		
Maximum Occurs On	Line Body		
Information			
Time	1. s		
Load Step	1		
Substep	1		
Iteration Number	1		

FIGURE 17
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 3 > Direct Stress

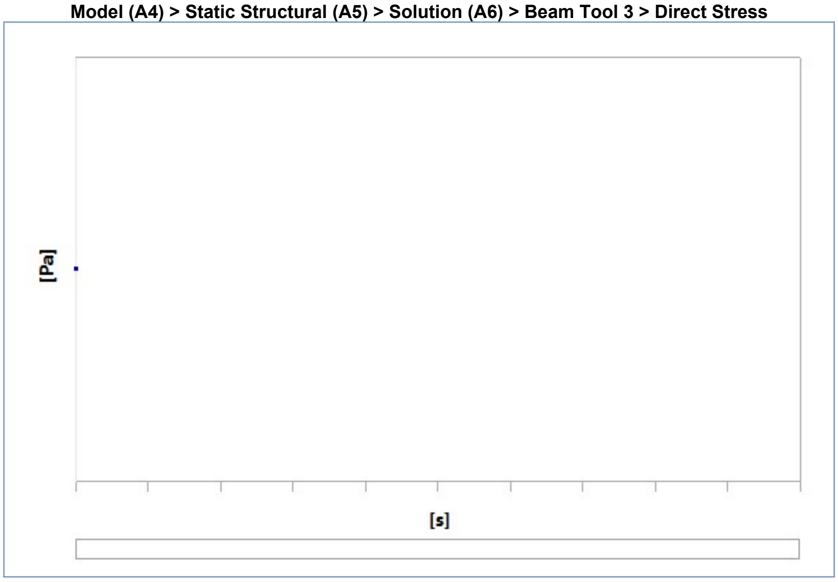


TABLE 30
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 3 > Direct Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	2.0388e+008	2.0388e+008	2.0388e+008

FIGURE 18
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 3 > Minimum Combined Stress

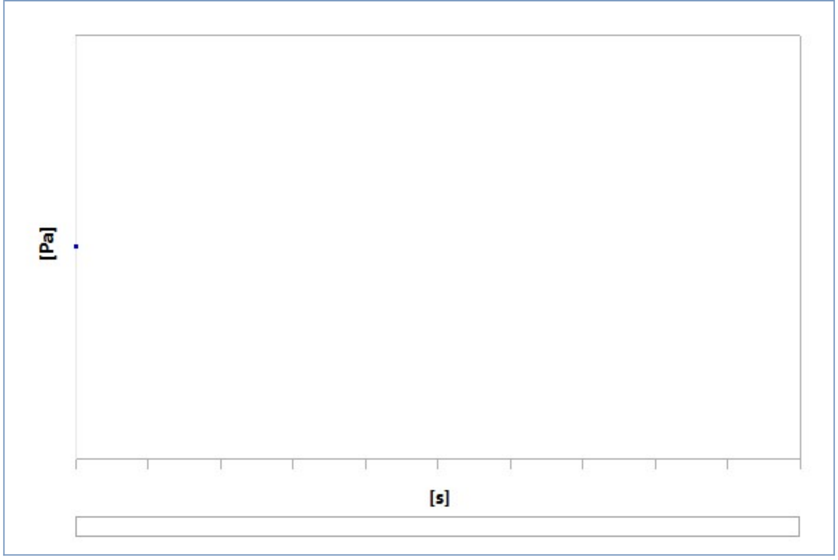


TABLE 31
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 3 > Minimum Combined Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	2.0388e+008	2.0388e+008	2.0388e+008

FIGURE 19
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 3 > Maximum Combined Stress

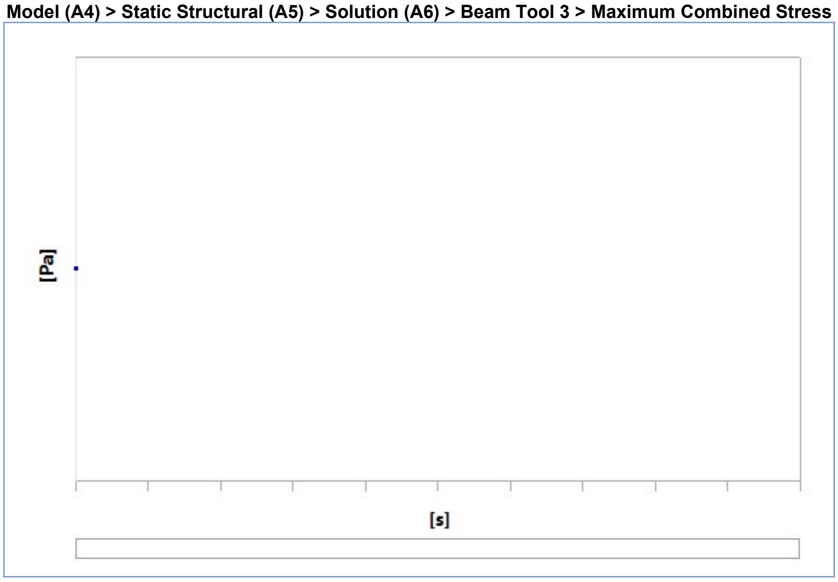


TABLE 32
Model (A4) > Static Structural (A5) > Solution (A6) > Beam Tool 3 > Maximum Combined Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	2.0388e+008	2.0388e+008	2.0388e+008

Material Data

Structural Steel

TABLE 33
Structural Steel > Constants

Density	7850 kg m^-3
Coefficient of Thermal Expansion	1.2e-005 C^-1
Specific Heat	434 J kg^-1 C^-1
Thermal Conductivity	60.5 W m^-1 C^-1
Resistivity	1.7e-007 kg m^3 A^-2 s^-3

TABLE 34
Structural Steel > Color

Red	Green	Blue
132	139	179

TABLE 35
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength Pa
0

TABLE 36
Structural Steel > Compressive Yield Strength

Compressive Yield Strength Pa
2.5e+008

TABLE 37
Structural Steel > Tensile Yield Strength

Tensile Yield Strength Pa
2.5e+008

TABLE 38
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
4.6e+008

TABLE 39
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 40
Structural Steel > S-N Curve

Alternating Stress Pa	Cycles	Mean Stress Pa
3.999e+009	10	0
2.827e+009	20	0
1.896e+009	50	0
1.413e+009	100	0
1.069e+009	200	0
4.41e+008	2000	0
2.62e+008	10000	0
2.14e+008	20000	0
1.38e+008	1.e+005	0
1.14e+008	2.e+005	0
8.62e+007	1.e+006	0

TABLE 41
Structural Steel > Strain-Life Parameters

Strength Coefficient Pa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient Pa	Cyclic Strain Hardening Exponent
9.2e+008	-0.106	0.213	-0.47	1.e+009	0.2

TABLE 42
Structural Steel > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
2.e+011	0.3	1.6667e+011	7.6923e+010	

TABLE 43
Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000